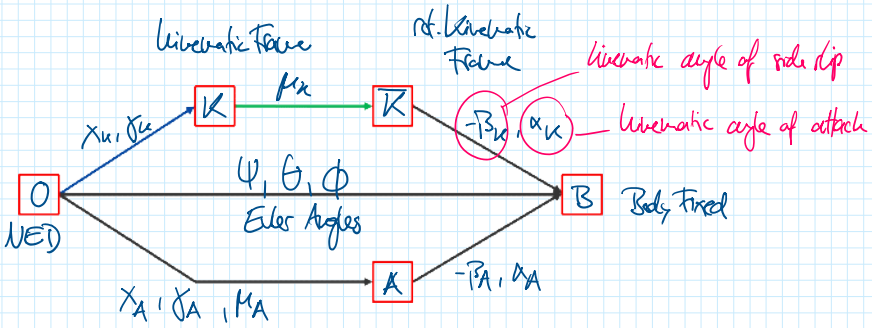
Exercise 4

Donnerstag, 20. November 2025 09:03



Exercise 1: Pointmass-Equations and Forces

A 1-engine propeller aircraft (piston engine) is flying in a windfree standard ISA-atmosphere without a bank angle in a straight flight with a climb angle γ_K of 0.5°. You are given the following data.

GPS-Data:

Geodetic Longitude	λ	11.669284 deg
Geodetic Latitude	μ	48.265906 deg
GPS-Height	h	5000 m
Climb angle	γ_K	0,5°

Propulsion-Data:

Reference thrust at MSL	T_i	5000 N
Reference density at MSL	ρ_i	$\rho_{s,MSL}$
Reference velocity	V_{ref}	80 kts
Density coefficient	n_ρ	0.75

Aircraft-Data:

Lift coefficient	C_L	1
Zero-lift drag coefficient	$C_{D,0}$	0,012
K-factor	k	0,016
Reference area	S_{ref}	20 m^2
Aircraft mass	m	2000 Kg

In the following you can assume that no aerodynamic side-forces act on the aircraft and that the kinematic bank angle μ_K equals the euler bank angle ϕ.

a) What is the required GPS velocity such that the z-component of the total force in the kinematic frame is zero?

$(F_{T,z})_K = 0 \rightarrow$ Corresponding velocity V

$$(\vec{F}_T)_K = (\vec{F}_A^A)_K + (\vec{F}_P^G)_K + (\vec{F}_G^G)_K$$

I. State all Forces in their "Easy" Frame which has no cross coupling

1) Aerodynamic Force $(\vec{F}_A^A)_A$

$$(\vec{F}_A^A)_A = \begin{pmatrix} X_A^A \\ Y_A^A \\ Z_A^A \end{pmatrix} = \begin{pmatrix} -D \\ Q \\ -L \end{pmatrix}_A = \bar{q} S \begin{pmatrix} -C_D \\ C_Q \\ -C_L \end{pmatrix}_A$$

2) Propulsive Force $(\vec{F}_P^G)_K$

$$(\vec{F}_P^G)_K = \begin{pmatrix} T \\ 0 \\ 0 \end{pmatrix}_K$$

$$\left(\vec{F}^{G}_{P}\right)_K = \begin{pmatrix} 0 \\ 0 \end{pmatrix}_K$$

3) Gravitational Force $\left(\vec{F}^{G}_{G}\right)_K$

$$\left(\vec{F}^{G}_{G}\right)_0 = \begin{pmatrix} 0 \\ 0 \\ mg \end{pmatrix}_0 = mg \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}_0$$

II. Transform into the desired frame

1) Gravitational Force

$$\left(\vec{F}^{G}_{G}\right)_K = M_{K0} \cdot \left(\vec{F}^{G}_{G}\right)_0$$

$$M_{K0} = \begin{bmatrix} \cos\gamma_K & 0 & -\sin\gamma_K \\ 0 & 1 & 0 \\ \sin\gamma_K & 0 & \cos\gamma_K \end{bmatrix} \begin{bmatrix} \cos\chi_K & \sin\chi_K & 0 \\ -\sin\chi_K & \cos\chi_K & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Rotation: γ_K, y-axis; 1. Rotation: χ_K, z-axis

$$\left(\vec{F}^{G}_{G}\right)_K = \begin{bmatrix} \cos\gamma_K\cos\chi_K & \sin\chi_K\cos\gamma_K & -\sin\gamma_K \\ -\sin\chi_K & \cos\chi_K & 0 \\ \cos\chi_K\sin\gamma_K & \sin\chi_K\sin\gamma_K & \cos\gamma_K \end{bmatrix} \cdot mg \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}_0 = \begin{pmatrix} -\sin\gamma_K \cdot mg \\ 0 \\ \cos\gamma_K \cdot mg \end{pmatrix}_K$$

2) Aerodynamic Force

$$\text{we used} \rightarrow \left(\vec{F}^{G}_{A}\right)_A = \left(\vec{F}^{G}_{A}\right)_{\bar{K}}$$

$$\left(\vec{F}^{G}_{A}\right)_K = M_{K\bar{K}} \cdot \begin{pmatrix} -D \\ Q \\ -L \end{pmatrix}_{A=\bar{K}}$$

$$\left(\vec{F}^{G}_{A}\right)_K = \begin{pmatrix} -D \\ L\sin\mu \\ -L\cos\mu \end{pmatrix}_K = \begin{pmatrix} -D \\ 0 \\ -L \end{pmatrix}_K$$

$\mu_K \doteq 0$

Rotation around x with μ_K

$$M_{K\bar{K}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\mu_K & -\sin\mu_K \\ 0 & \sin\mu_K & \cos\mu_K \end{bmatrix}$$

switched sign because we go from $\bar{K} \rightarrow K$

III. Summarize all Forces

$$\left(\vec{F}^{G}_{T}\right)_K = \begin{pmatrix} -D \\ 0 \\ -L \end{pmatrix}_K + \begin{pmatrix} T \\ 0 \\ 0 \end{pmatrix}_K + \begin{pmatrix} -mg\sin\gamma_K \\ 0 \\ mg\cos\gamma_K \end{pmatrix}_K$$

IV. Determine the Velocity $V = V_{TAS}$ for the z-component to be zero

$$0 = -L + 0 + mg \cdot \cos\gamma_K$$

$$0 = -\frac{1}{2}\rho S V^2 \cdot c_L + mg \cdot \cos\gamma_K$$

$$\frac{1}{2}\rho V^2 \cdot S_{ref} \cdot c_L = mg \cdot \cos\gamma_K$$

$$V_{TAS} = \sqrt{\frac{2mg\cos\gamma_K}{\rho S_{ref} \cdot c_L}} = 51{,}82\ \frac{m}{s};$$

ρ → unknown! But ISA-conditions

ISA:

$$\frac{\rho}{\rho_{SL}} = \left(1 + \frac{\gamma_{TR}}{T_R} \cdot H_G\right)^{\frac{1}{\mu_{TR}-1}}$$

$$\rho = \rho_{SL}\left(1 + \frac{\gamma_{TR}}{T_R} \cdot H_G\right)^{\frac{1}{\mu_{TR}-1}} = 0.736\ \frac{kg}{m^3};$$

H_G → $H_G = \frac{\varepsilon \cdot h}{\varepsilon + h} = 4996\,m$

b) Calculate the total force acting on the aircraft. State the result in a suitable coordinate frame.

$$\left(\vec{F}_{T}^{\,a}\right)_K = \begin{pmatrix} -mg\sin\gamma_a \\ 0 \\ mg\cos\gamma_a \end{pmatrix}_K + \begin{pmatrix} -D \\ 0 \\ -L \end{pmatrix}_K + \begin{pmatrix} T \\ 0 \\ 0 \end{pmatrix}_K$$

already known as 0 (see subtask a)

given — unknown but V_{TAS} from a — unknown

Drag: $D = \frac{1}{2}\rho V^2 S_{ref} \cdot c_D = 549.16\,N$

$c_D = c_{D0} + k \cdot c_L^2 = 0.028$

Thrust: $\frac{T}{T_i} = \left(\frac{V}{V_i}\right)^{n_V} \cdot \left(\frac{\rho}{\rho_i}\right)^{n_\rho}$ given: Propeller aircraft $\Rightarrow n_V = -1$

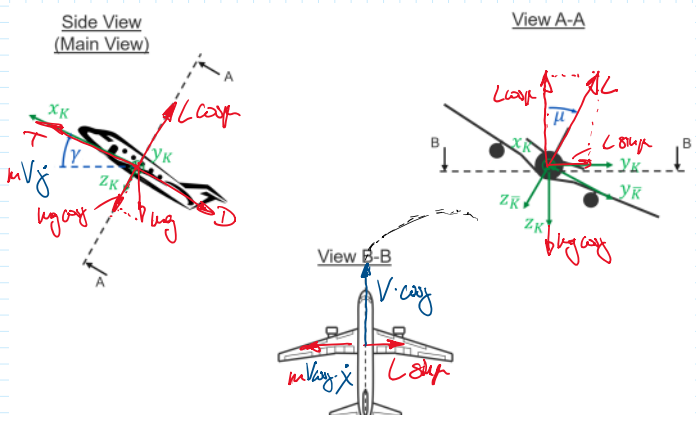
$$T = T_i \cdot \left(\frac{V}{V_i}\right)^{n_V} \cdot \left(\frac{\rho}{\rho_i}\right)^{n_\rho} = 2720.19\,N$$

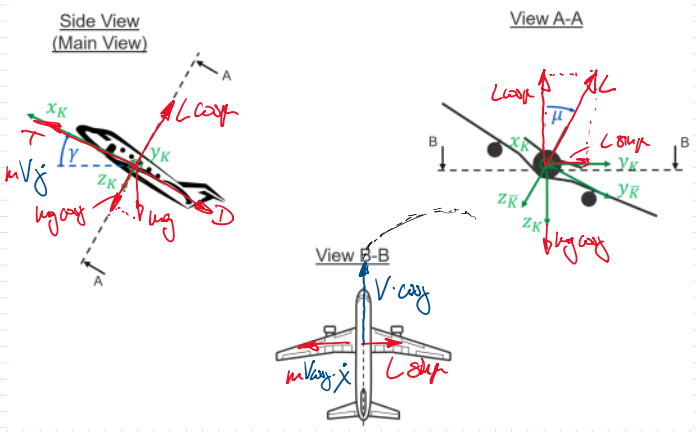
$\Rightarrow$ Complete the System: $\left(\vec{F}_{T,x}^{\,a}\right)_K = T - D - mg \cdot \sin\gamma_a = 1999.88\,N$;

$\Rightarrow \left(\vec{F}_{T}^{\,a}\right)_K = \begin{pmatrix} 1999.88\,N \\ 0\,N \\ 0\,N \end{pmatrix}_K$

c) Calculate the current time derivative of the aircraft states.

States: $V, \chi, \gamma \rightarrow \dot{V}, \dot{\chi}, \dot{\gamma}$





$$\dot{V} = \frac{T-D}{m} - g\sin\gamma = 0.999\ m/s^2$$

$$L = \frac{1}{2}\rho V^2 S_{ref} \cdot c_L$$

$$\dot{\chi} = \frac{L\sin\mu_K}{mV\cos\gamma} = 0;$$

$$\dot{\gamma} = \frac{L\cos\mu_K}{mV} - \frac{g}{V}\cos\gamma = 0;$$

d) What happens to the time derivatives if the aircraft flies with a kinematic bank angle $\mu_K = 30°$ if all other states remain the same?

$$\dot{V} = \frac{T-D}{m} - g\cdot\sin\left(0.5°\cdot\frac{\pi}{180}\right) = 0.999\ m/s^2;$$

$$\dot{\chi} = \frac{L\cdot\sin\left(30°\cdot\frac{\pi}{180}\right)}{mV\cos\left(0.5\cdot\frac{\pi}{180}\right)} = 0.095\ rad/s;$$

$$\dot{\gamma} = \frac{L\cdot\cos\left(30\frac{\pi}{180}\right)}{mV} - \frac{g}{V}\cos\left(0.5\cdot\frac{\pi}{180}\right) = -0.0295\ rad/s;$$

$L^* < G \Rightarrow$ Sink;

